Exercise

I.Chapter3: Electricity and magnetism

A. Charges, atoms and electrons

1. How many types of electric charges? List it down.
2. State the law of repulsion and attraction of electric charge?
3. Why the polythene stirp can attract cellulose acetate? Explain.
4. What is the inside of an atom? How many particles contain in an atom? List it down.
5. When an atom becomes electrically neutral and electrically charged?
6. What is the difference between conductor and insulator?
7. When the electrostatic occur? Explain. What is the electric field?
8. What is the device used for prevent tall building from the lightning? How does it work? Explain.

II.Chapter3: Electric current

B. Effect of electric current

9. What is the direction of positive ions and negative ions flow in electric circuit?
10. What is the direction of electric current flow in electric circuit?
11. What are the effects of electric current flow in electric circuit? List it down.
12. What is the flow of charge called?
13. **(a)** In which direction does conventional current flow around a circuit?
    (b) In which direction do electrons flow?
14. Which of the following equation shows the correct relationship between electrical units?
    (a) 1 A = 1 C/s or **(b)** 1 C = 1 A/s
15. What is the unit of electric current?
16. **(a)** How many milliamperes are there in 1 ampere?
    (b) How many microamperes are there in 1 ampere?
17. **(a)** What is an electric current? What makes an electric current flow in a wire?
    (b) Define the unit of electric current (or Define ampere).

C. Ampere and Coulomb

18. How many types of electric circuit? List it down.
19. How many types of electric current? What is the difference between d.c and a.c current? Draw graphic of d.c and a.c current respect with time?
20. What is the frequency of alternating current? How the frequency of alternating current work?
21. If 20 C of charge pass a point in a circuit in 1 s, what current is flowing?
22. A current of 4 A flows around a circuit for 10 s. How much charge flows past a point in the circuit in this time?
23. A component has a current of 0.430 A passing through it. Calculate the charge passing through the component in exactly 3 minutes.
24. What is the current in a circuit if the charge passing each point is 20 C in 40 s?
25. If the charge on an electron is 1.6×10^{-19} coulombs, how many electrons should pass through a conductor in 1 second to constitute 1 ampere current?
26. The p.d. across a lamp is 12 V. How many joules of electrical energy are changed into heat and light when:
    (a) a charge of 1 C passes through it?
    (b) a charge of 5 C passes through it?
    (c) a current of 2 A flows through it for 10 s?
27. In 10 s, a charge of 25 C leaves a battery, and 200 J of energy are delivered to an outside circuit as a result.
    (a) What is the p.d. across the battery?
    (b) What current flows from the battery?
28. **(a)** Define electric current. What is the SI unit of electric current?
    (b) One coulomb of charge flows through any cross-section of a conductor in 1 second. What is the current flowing through the conductor?
    (c) Which instrument is used to measure electric current? How should it be connected in a circuit?

(d) What is the conventional direction of the flow of electric current? How does it differ from the direction of flow of electrons?

(e) A flash of lightning carries 10 C of charge which flows for 0.01 s. What is the current? If the voltage is 10 MV, what is the energy?

29. Which unit could be used to measure current?

(a) Watt **(b)** Coulomb **(c)** Volt **(d)** Ampere

30. If the current through a floodlamp is 5 A, what charge passes in 10 seconds?

(a) 0.5 C **(b)** 2 C **(c)** 5 C **(d)** 50 C

31. If the amount of electric charge passing through a conductor in 10 minutes is 300 C, the current flowing is:

(a) 30 A **(b)** 0.3 A **(c)** 0.5 A **(d)** 5 A

32. Draw a circuit diagram to show how 3 bulbs can be lit from a battery so that 2 bulbs are controlled by the same switch while the third bulb has its own switch.

33. An electric heater is connected to the 230 V mains supply. A current of 8 A flows through the heater.

(a) How much charge flows around the circuit each second?

(b) How much energy is transferred to the heater each second?

34. How many electrons are flowing per second past a point in a circuit in which there is a current of 5 amp?

D. Mean drift velocity

35. A particle has a charge of 4e. What is the charge of the particle in coulombs?

36. A cell has a charge of 91 C passing through it every 32 seconds. Calculate the current passing through the cell.

37. A copper wire has a cross-sectional area of 0.50×10^{-6} m^2. The number density of electrons for copper is 8.5×10^{28} m^{-3}. A current of 12 A flows through the wire. Calculate the mean drift velocity of the free electrons in the wire.

38. Here you do 6 J of work moving each coulomb of charge through the resistor, so the p.d. across it is 6 V.

E. Chpater3: Potential difference

39. Define the term current.

40. Are ammeters attached to a component in series or in parallel?

41. Define the term potential difference.

42. In a circuit, it takes 114 J to move 56.0 C of charge through a filament lamp. Calculate the potential difference across the lamp.

43. A resistor has a current of 1.30 A passing through it.

a) Calculate the charge that passes through it in exactly 12 minutes.

b) The potential difference across the resistor is 24.0 V. Calculate the total work done to move the charge through the resistor over 12.0 minutes.

F. Ohm's law

44. What is meant by resistance?

Ans: Resistance is the measure of the opposition to the current flow in the electrical circuit.

45. What is Ohm's law?

Ans: Ohm's law state that the resistance of a conductor when the current 1 ampere passed through it due to voltage 1 volt has been applied across it.

46. What happens to the resistance of an ohmic conductor if you double the potential difference across it? Give a reason for your answer.

Ans: Nothing affect to the resistance of a conductor due to double voltage applied across it.

47. A potential difference of 12 V is applied across a resistor of 240 Ω. Calculate the current through the resistor.

Ans: $\mathbf{I = \frac{V}{R} = \frac{12}{240} = 0.05A}$

48. The potential difference across a component is 230 V. The current flowing through the component is 12.4 A. Calculate the resistance of the component.

Ans: $\mathbf{R = \frac{V}{I} = \frac{230}{12.4} = 18.54\Omega}$

49. A fixed resistor has a resistance of 4.43 Ω and the potential difference across it is 12.0 V. Calculate the current flowing through the fixed resistor.

Ans: $\mathbf{I = \frac{V}{R} = \frac{12}{4.43} = 2.71A}$

50. A thermistor has a resistance of 8.62 Ω. Calculate the potential difference across the thermistor when there is a current of 2.10 A flowing through it.

Ans: $\mathbf{V = RI = (8.62)(2.10) = 18V}$

51. A fuse has a current of 1.21 A flowing through it. Calculate the resistance of the fuse when the potential difference across it is 13.4 V.

Ans: $\mathbf{R = \frac{V}{I} = \frac{13.4}{1.21} = 11\Omega}$

52. A student is testing a component in a circuit by varying the current in the circuit and measuring the voltage across the component. As the current increases, the voltage stays the same. What is happening to the resistance of the component?

Ans: The resistance is decreases as the current increase while voltage keeps constant because the resistance is inversely proportional to the current flow in the electrical circuit.

53. A section of copper wire acts as an ohmic conductor. When the current flowing through the wire is 3.20 A, the potential difference across the wire is 15.6 V. Calculate the potential difference across the wire when the current is increased to 4.10 A.

Ans: $\mathbf{V = RI}$ **but** $\mathbf{R = \frac{V}{I} = \frac{15.6}{3.20} = 4.88\Omega}$

Imply $\mathbf{V = (4.88)(4.10) = 20V}$

54. The current flowing through a resistor is 0.8 A when a p.d. of 20 V is applied. Determine the value of the resistance.

Ans: $\mathbf{R = \frac{V}{I} = \frac{20}{0.8} = 25\Omega}$

55. Determine the p.d. which must be applied to a 2 kΩ resistor in order that a current of 10 mA may flow.

Ans: $\mathbf{V = RI = (2000)(10 \times 10^{-3}) = 20000 \times 10^{-3}V = 20V}$

56. A coil has a current of 50 mA flowing through it when the applied voltage is 12 V. What is the resistance of the coil?

Ans: $\mathbf{R = \frac{V}{I} = \frac{12}{50\times10^{-3}} = 0.24 \times 10^{3}\Omega = 240\Omega}$

57. A 100 V battery is connected across a resistor and causes a current of 5 mA to flow. Determine the resistance of the resistor. If the voltage is now reduced to 25 V, what will be the new value of the current flowing?

Ans: $\mathbf{R = \frac{V}{I} = \frac{100}{5\times10^{-3}} = 20 \times 10^{3}\Omega = 20k\Omega}$

$\mathbf{I = \frac{V}{R} = \frac{25}{20\times10^{3}} = 1.25 \times 10^{-3} = 1.25mA}$

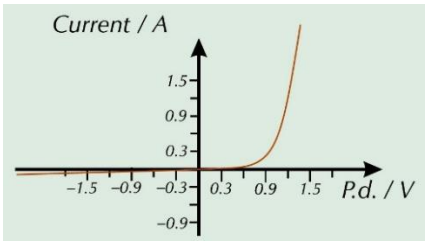
58. What is the resistance of a coil which draws a current of (a) 50 mA and (b) 200 μA from a 120 V supply?

Ans: (a). $\mathbf{R = \frac{V}{I} = \frac{120}{50\times10^{-3}} = 2.4 \times 10^{3}\Omega = 2400\Omega}$

(b). $\mathbf{R = \frac{V}{I} = \frac{120}{200\times10^{-6}} = 0.6 \times 10^{6}\Omega = 600k\Omega}$

G. Semiconductor diode, ohmic conductor and non-ohmic conductor

59. The graph below shows the characteristic I-V curve for a component.



a) What type of component has this I-V characteristic graph?

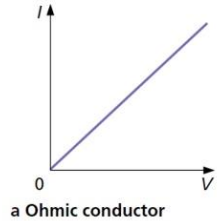
b) Describe how the current and voltage vary in the forward bias direction.

c) Describe the resistance of the component in the reverse direction.

Ans: (a). Semiconductor diode.

(b). The diode requires a voltage of about 0.6 V in the forward direction before it will conduct, so there is no current flow until 0.6 V on the diagram. After 0.6 V, the current is allowed to flow, and increases (exponentially) with an increase in potential difference.

(c). Resistance is very high in reverse bias. R=V/I=1.5/-0=INFINITY



a Ohmic conductor

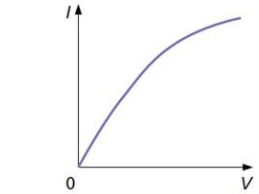
60. What resistances do you normally assume a voltmeter and an ammeter to have?

Ans: The resistance should locate at the forward bias in order to allow Voltmeter and ammeter to measure.

61. Sketch the characteristic I-V graph of:

a) an ohmic conductor under constant physical conditions,

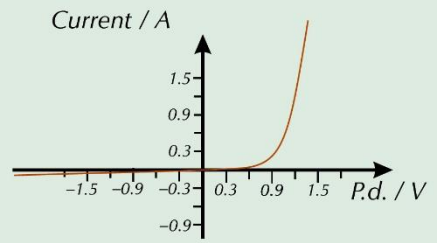
b) a filament lamp.



b Filament lamp

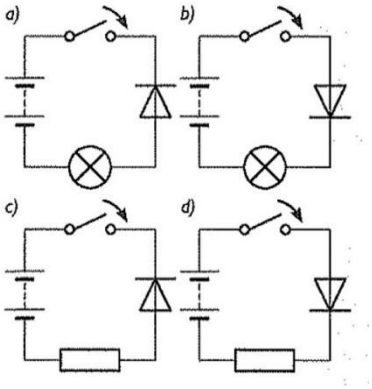
Figure 38.7 *I–V* graphs

62. Sketch the characteristic V-I graph of a diode.



63. Is a filament lamp an ohmic conductor? How can you tell from its characteristic I-V graph?
Ans: No, It's not. From the graph of filament, the current pass through the non-ohmic conductor is increases as voltage or potential difference increases.

64. State what happens in each of the following circuits when the switch is closed.



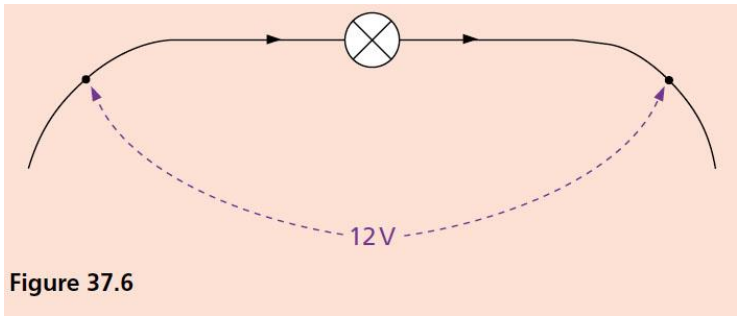
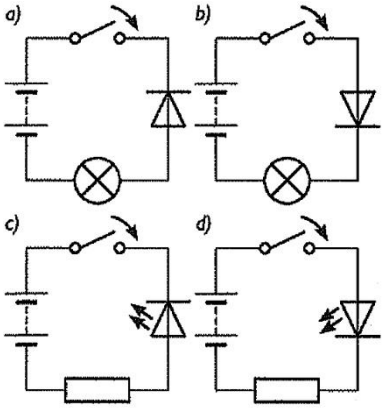


Figure 37.6

Ans: (Figure left-hand side) statement of diode circuit



a) The lamp **not** light.
b) The lamp **does** light.
c) The LED does **not** light.
d) The LED **does** light.
Note: LEDs are destroyed if too much current passes through them. The resistor in series with the LED is needed to limit the current.

(Figure right-hand side), when the switch is off, then the lamp is light up.

65. The p.d. across the lamp in Figure 37.6 is 12 V. How many joules of electrical energy are changed into light and heat when
a a charge of 1 C passes through it, **Ans:** $\mathbf{E = QV = (1C)(12V) = 12J}$
b a charge of 5 C passes through it, **Ans:** $\mathbf{E = QV = (5C)(12V) = 60J}$
c a current of 2 A flows in it for 10 s?
Ans: $\mathbf{E = QV = ItV = (2A)(10s)(12V) = 240J}$

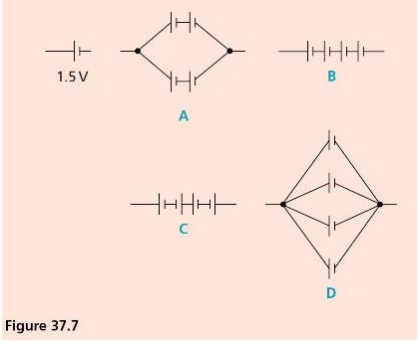
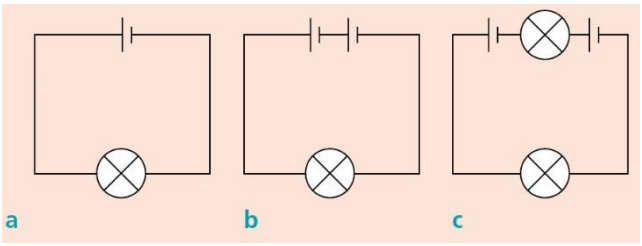


Figure 37.7

66. Three 2 V cells are connected in series and used as the supply for a circuit.
a What is the p.d. at the terminals of the supply?
b How many joules of electrical energy does 1 C gain on passing through
(i) one cell, **(ii)** all three cells?
(a). Ans: $\mathbf{V = \varepsilon = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 2V + 2V + 2V = 6V}$
(b). i. $\mathbf{E = QV = (1)(2V) = 2J}$ **(ii).** $\mathbf{E = QV = (1C)(6V) = 6J}$

67. Each of the cells shown in Figure 37.7 has a p.d. of 1.5 V. b Which of the arrangements would produce a battery with a p.d. of 6 V?
Ans: The arrangement of circuit diagram (b) is the correct answer.

68. The lamps and the cells in all the circuits of Figure 37.8 are the same. If the lamp in **a** has its full, normal brightness, what can you say about the brightness of the lamps in **b**, **c**, **d**, **e** and **f**?



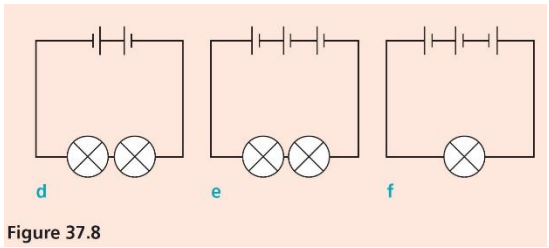


Figure 37.8

Ans: State the brightness of the lamps in circuit diagram b, c, d, e and f

b Very bright c Normal brightness d No light e Brighter than normal f Normal brightness

69. Three voltmeters V, V1 and V2 are connected as in Figure 37.9.

a If V reads 18 V and V1 reads 12 V, what does V2 read?

b If the ammeter A reads 0.5 A, how much electrical energy is changed to heat and light in lamp L1 in one minute?

c Copy Figure 37.9 and mark with a + the positive terminals of the ammeter and voltmeters for correct connection.

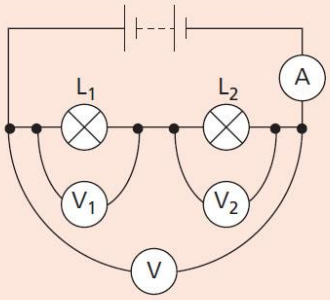


Figure 37.9

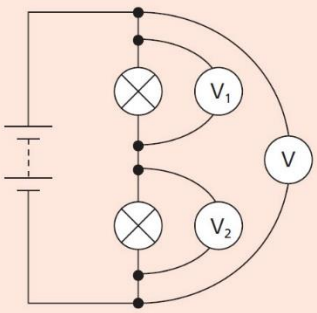


Figure 37.10

Ans: a) $\mathbf{V = V_1 + V_2 \Rightarrow V_2 = V - V_1 = 18V - 12V = 6V}$

b) $\mathbf{E = QV = ItV = (0.5A)(60s)(12V) = 360J}$

70. Three voltmeters are connected as in Figure 37.10.

What are the voltmeter readings x, y and z in the table below (which were obtained with three different batteries)?

V/V	V_1/V	V_2/V
x	12	6
6	4	y
12	z	4

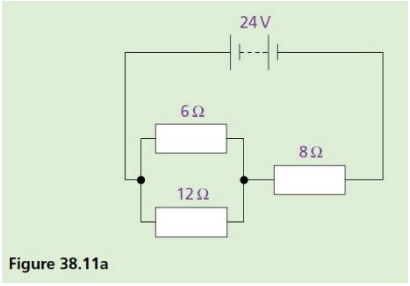


Figure 38.11a

Ans: $\mathbf{X = Y + Z = 6 + 12 = 18V}$

$\mathbf{Y = X - Z = 6 - 4 = 2V}$

$\mathbf{Z = X - Y = 12 - 4 = 8V}$

71. A p.d. of 24 V from a battery is applied to the network of resistors in Figure 38.11a.

a What is the combined resistance of the 6 Ω and 12 Ω resistors in parallel?

Ans: $\mathbf{\frac{1}{R_{12}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{1}{R_{12}} = \frac{1}{6} + \frac{1}{12} = \frac{1\times2}{6\times2} + \frac{1}{12} = \frac{3}{12} \Rightarrow R_{12} = \frac{12}{3} = 4\Omega}$

b What is the current in the 8 Ω resistor?

Ans: $\mathbf{V = R_3 I \Rightarrow I = \frac{V}{R_3} = \frac{24}{8} = 3A}$

c What is the voltage across the parallel network?

Ans: $\mathbf{V_1 = V_2 = R_{12} I = 4 \times 3 = 12V}$

d What is the current in the 6 Ω resistor?

Ans: $\mathbf{I_1 = \frac{V_1}{R_1} = \frac{12}{6} = 2A}$